

Bioinformatic approaches to regulatory genomics and epigenomics

376-1347-00L | week 11

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Plan for today

- Debriefing on the assignment
- Single-cell epigenomic data (scATAC-seq)
- Some lessons from multi-omic integration

Debriefing on the assignment

- Most distal interactions were within 100kb.
- The proportion of predicted distal targets matching the nearest target ranged from 10 to 35% (high for factors that bind chiefly at the promoter, and lower for those binding chiefly at distal enhancers).
- When the interactions are distal, whether it's downstream or upstream doesn't make a difference

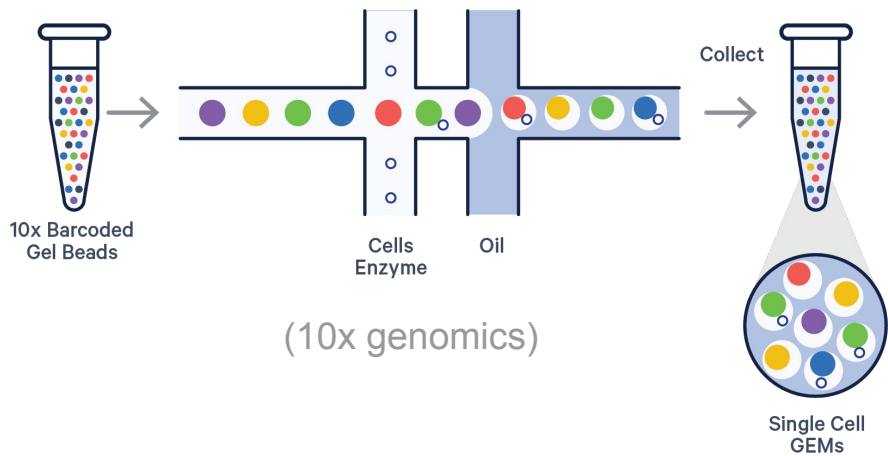
NOTE:

Most of you repeated peaks when they had more than one interaction. While this is legit, keep in mind that the expanded ranges are then not individual peaks anymore. As such, it would be wrong to say something like “X% of the peaks interact with the closest gene” (but a correct formulation would be “X% of the predicted interactions are with the closest gene”)

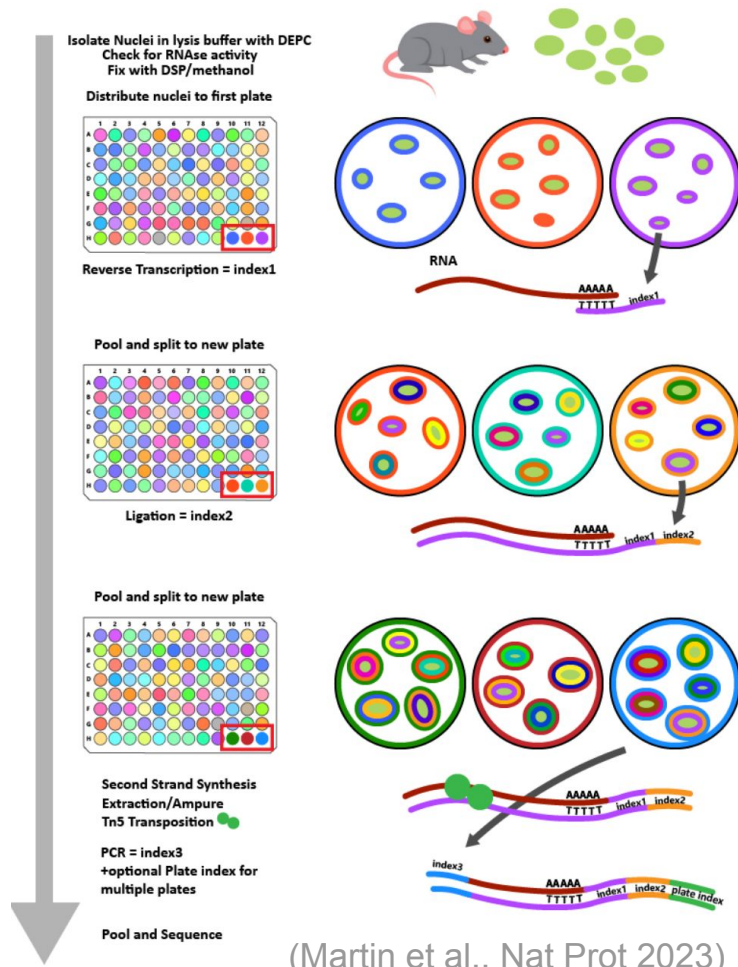
Single-cell *-omics

Two main families of approaches for single-cell sequencing methods:

Droplet-based (e.g. 10x)

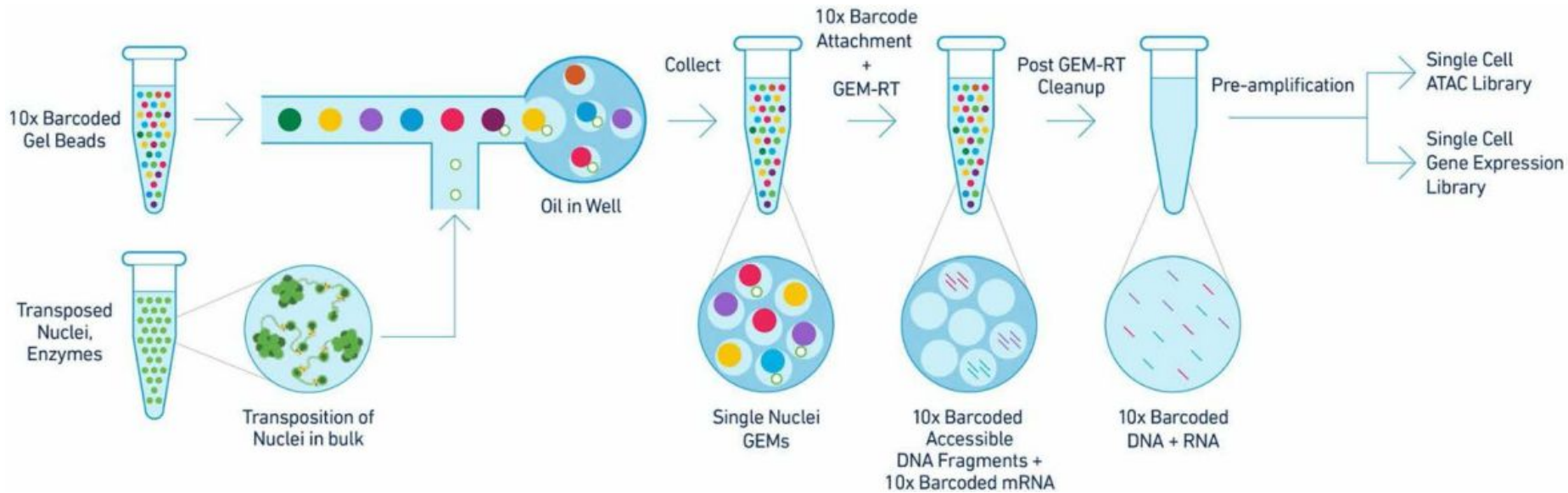


Combinatorial indexing

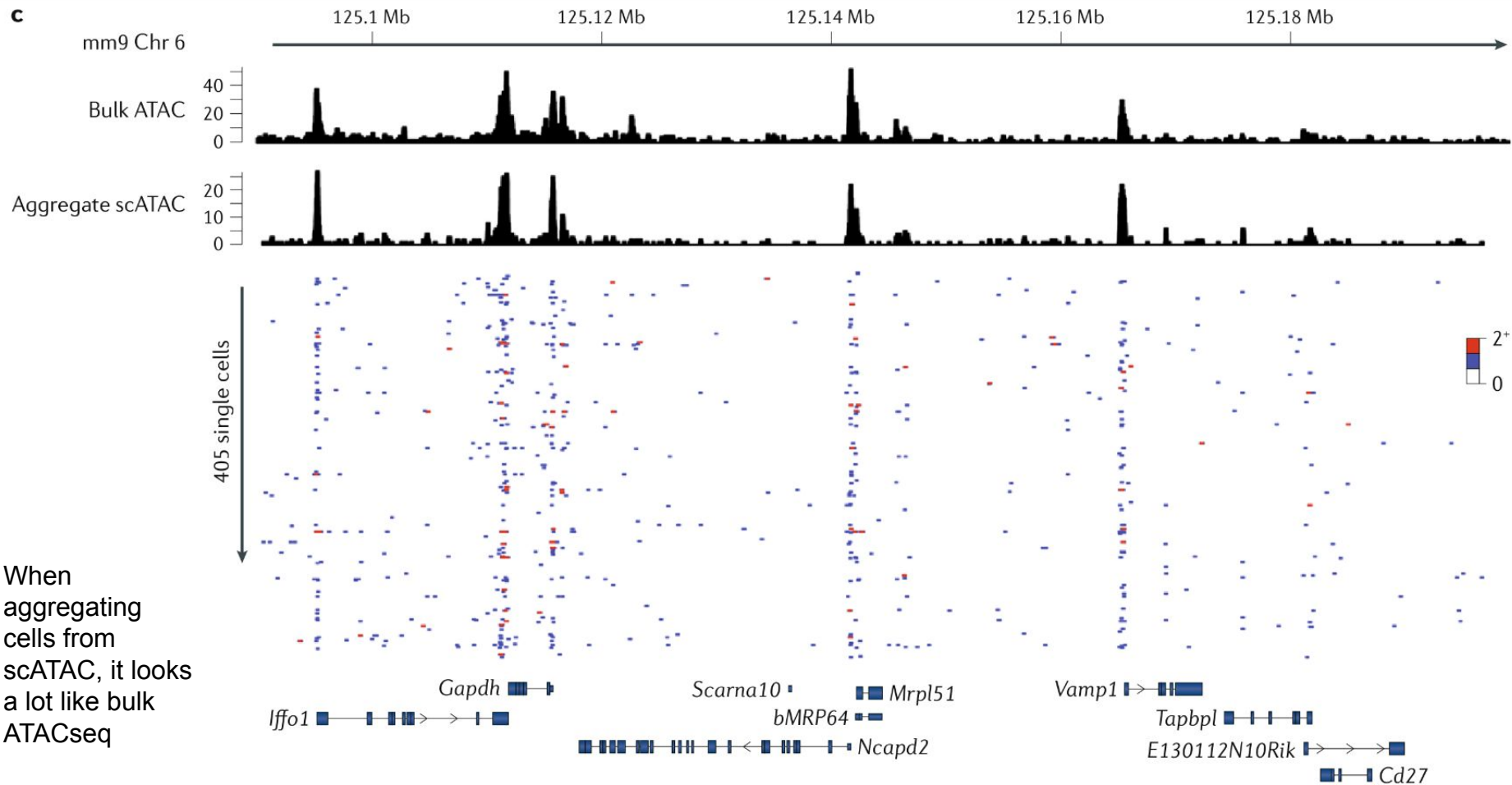


(Martin et al., Nat Prot 2023)

Single-cell ATAC-seq (and multi-omics)



(10x Genomics)



(Mezger et al, Nat Comm 2018)

Single-cell ATACseq analysis in a nutshell

1. The output of the genome alignment of the data is a “fragment file”, a bed-like file containing the coordinates of each fragment and the associated cell (barcode)

#chr	start	end	cell_barcode
chr1	10066	10536	TCAAGCAGTGCATC-1
chr1	10073	10278	TCAAGACGTCTGATTG-1
chr1	10073	10305	CGTTCCACAGCGTAGA-1
chr1	10079	10315	TTCAACTTCCGAGAGA-1
chr1	10085	10278	TCGTTGCATAGGCGA-1
chr1	10091	10303	AGCGTGCTCCCATAGA-1

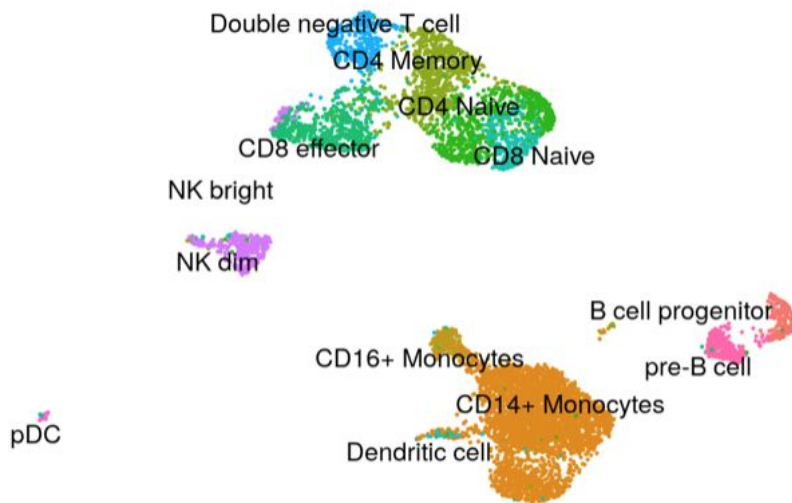
2. From this, we count the number of fragments from each cell overlapping genomic windows (either whole-genome tiles or feature-based)

	cell1	cell2	cell3	cell4	cell5	...
window1	1	0	0	0	0	
window2	0	0	0	0	1	
window3	0	0	1	0	0	
window4	0	0	0	0	0	
window5	0	1	0	0	0	
...						

Can be a matrix
with hundreds of
thousands
windows...

(filtering...)

3. Normalization, dimensional reduction (e.g. LSI), and clustering of the cells:

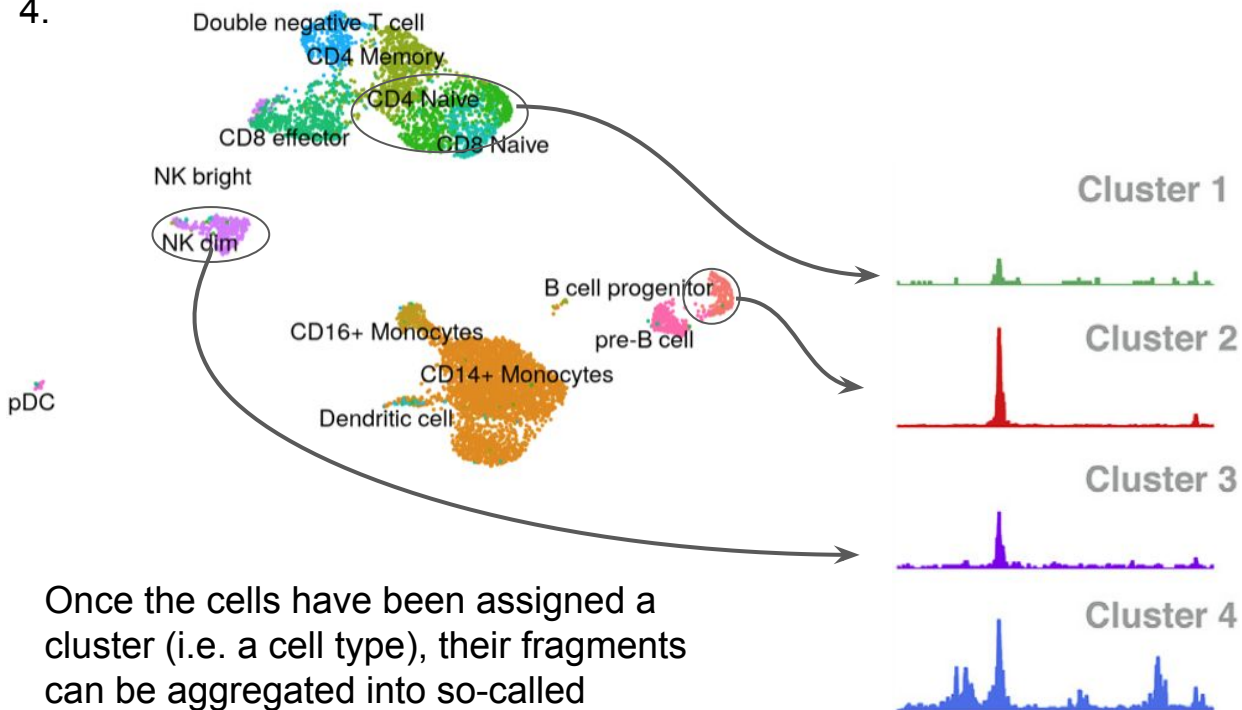


Four main pipelines:

- [Signac](#)
- [snapATAC2](#) (python)
- [ArchR](#) (feature-rich, scalable, but suboptimal – [ref](#))
- [BPCells](#) (most scalable)

Single-cell ATACseq analysis in a nutshell

4.



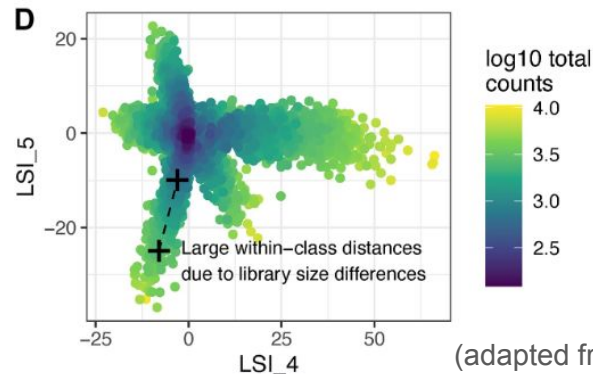
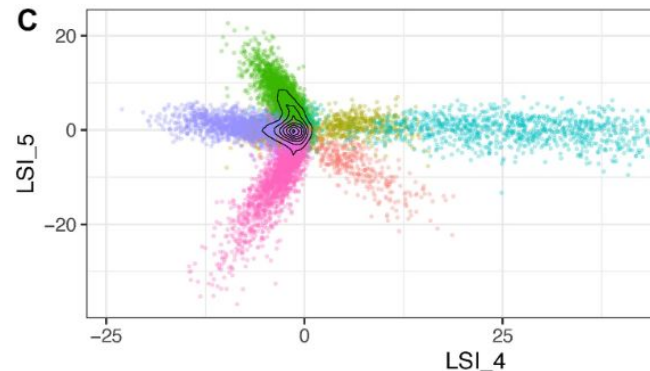
Once the cells have been assigned a cluster (i.e. a cell type), their fragments can be aggregated into so-called “**pseudo-bulk**” profiles

From this point on, the data is pretty much like traditional (bulk) ATACseq data, meaning that you can apply all the tools you’re familiar with, but it’s cell-type-specific!

One also often do some work at the pseudo-bulk level (e.g. calling peaks) before going back to the cell-level

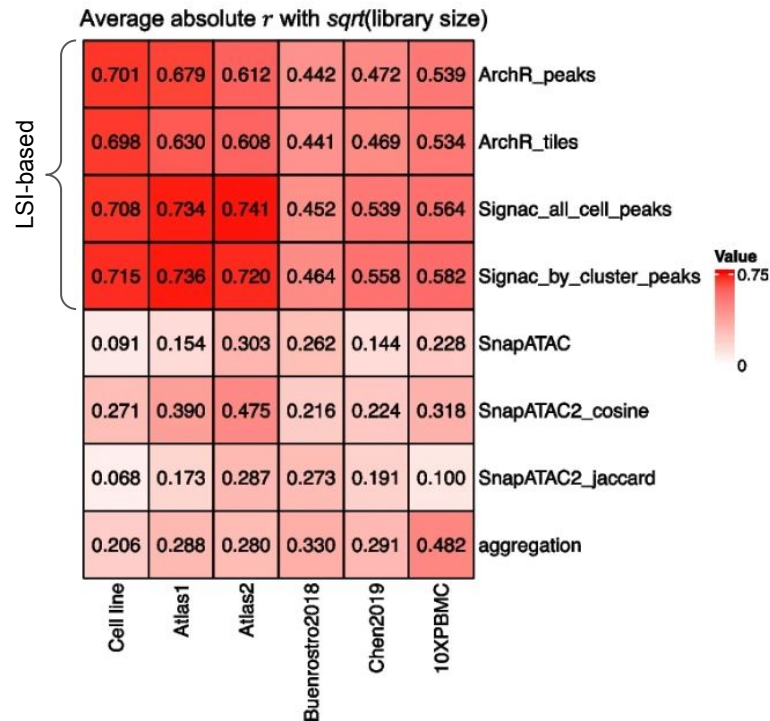
A warning on Latent Semantic Indexing (LSI) of snATAC

LSI (TF-IDF normalization followed by SVD) is the most widespread way to process single-cell epigenomic signal, but much of the variation in the embedding is driven by differences in library sizes (even after component filtering).



Using cosine distance solves much of the problem, but euclidean distances on LSI embeddings are a no-go!

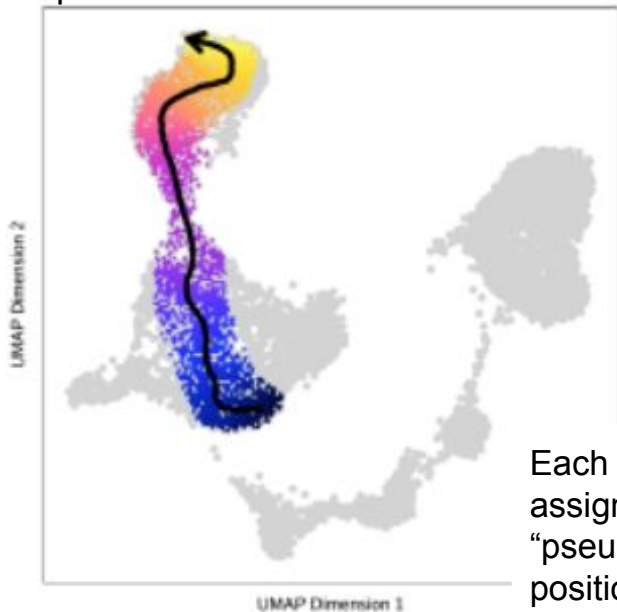
(adapted from Luo et al., NAR 2025)



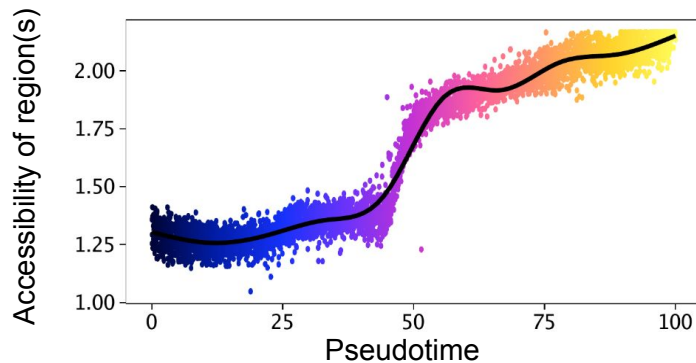
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Single-cell ATACseq analysis – doing more at the cell-level

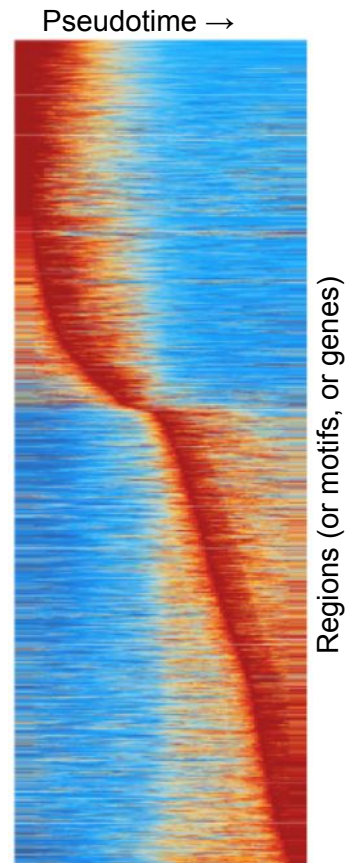
We identify “trajectories” across cells, representing for example a process of differentiation



Each cell can be assigned a “pseudotime”, i.e. it’s position along the trajectory



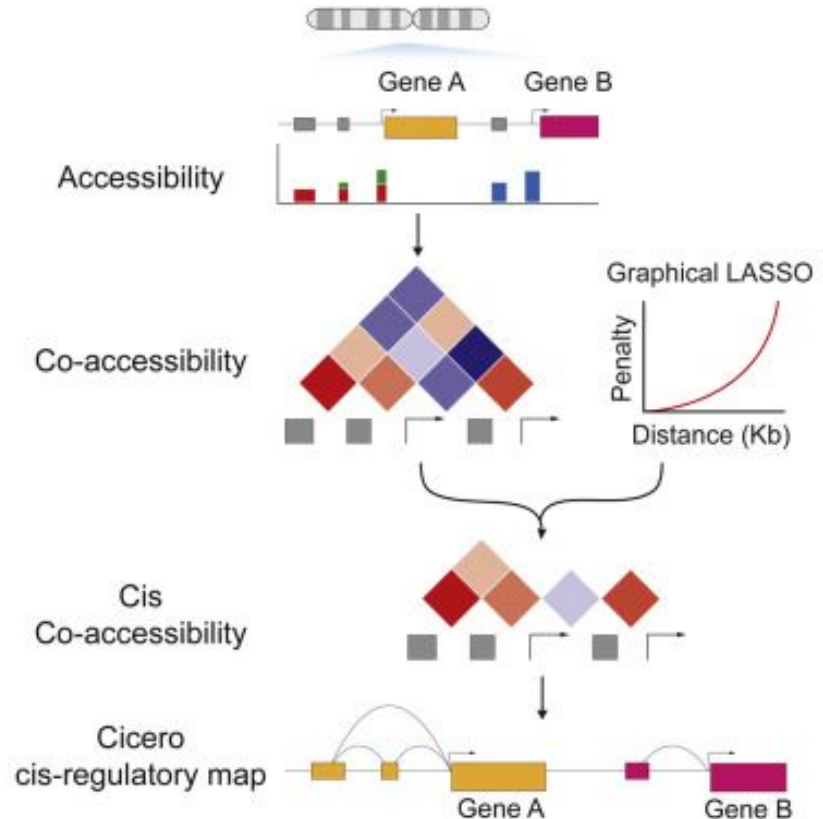
We can then track the accessibility of regions of interest across this “pseudotime”



Single-cell ATACseq analysis – doing more at the cell-level

Because we have so many cells, we can use the correlation between the accessibility at distal regulatory elements (i.e. enhancers) and putative TSS to know what genes these regulate

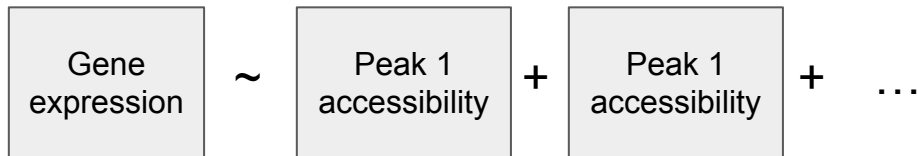
Due to the very high sparsity and noise of the data, this does not always work well. What does work well, however, is to first aggregate together groups of cells (*meta-cells*) that are highly similar, and then test correlations across meta-cells (Pliner et al., 2018; Persad et al., 2023)



Single-cell ATACseq analysis – doing more at the cell-level

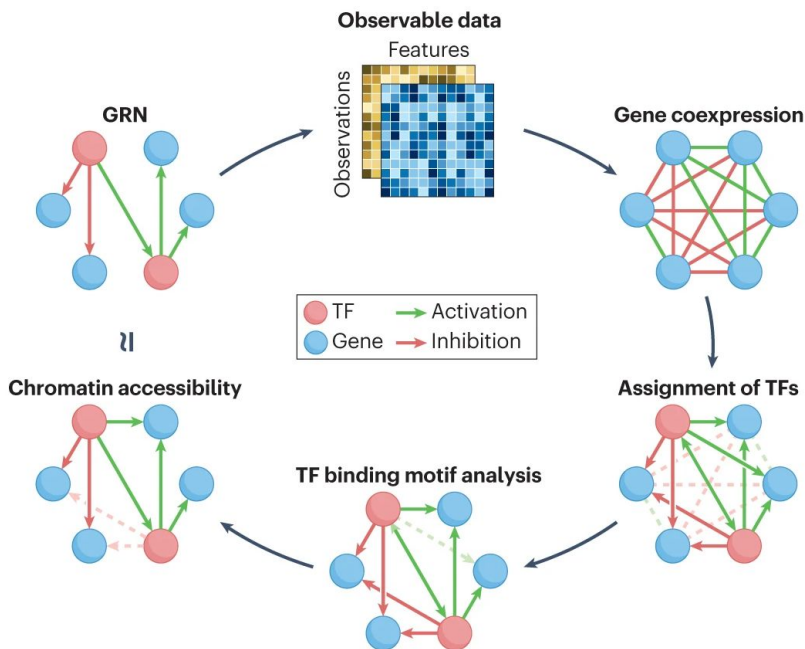
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Using multimodal (ATAC + RNA) data, predict:



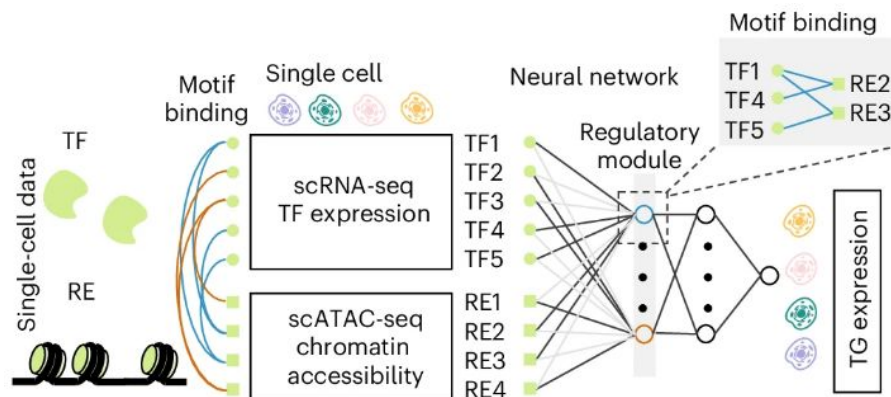
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Gene regulatory network inference



(adapted from Badia-i-Mompel et al., *Nat Rev Gen* 2023)

Example method : LINGER



(adapted from Yuan & Duren, *Nat Bio* 2024)

See especially:

[SCENIC+](#) and [epiregulon](#)

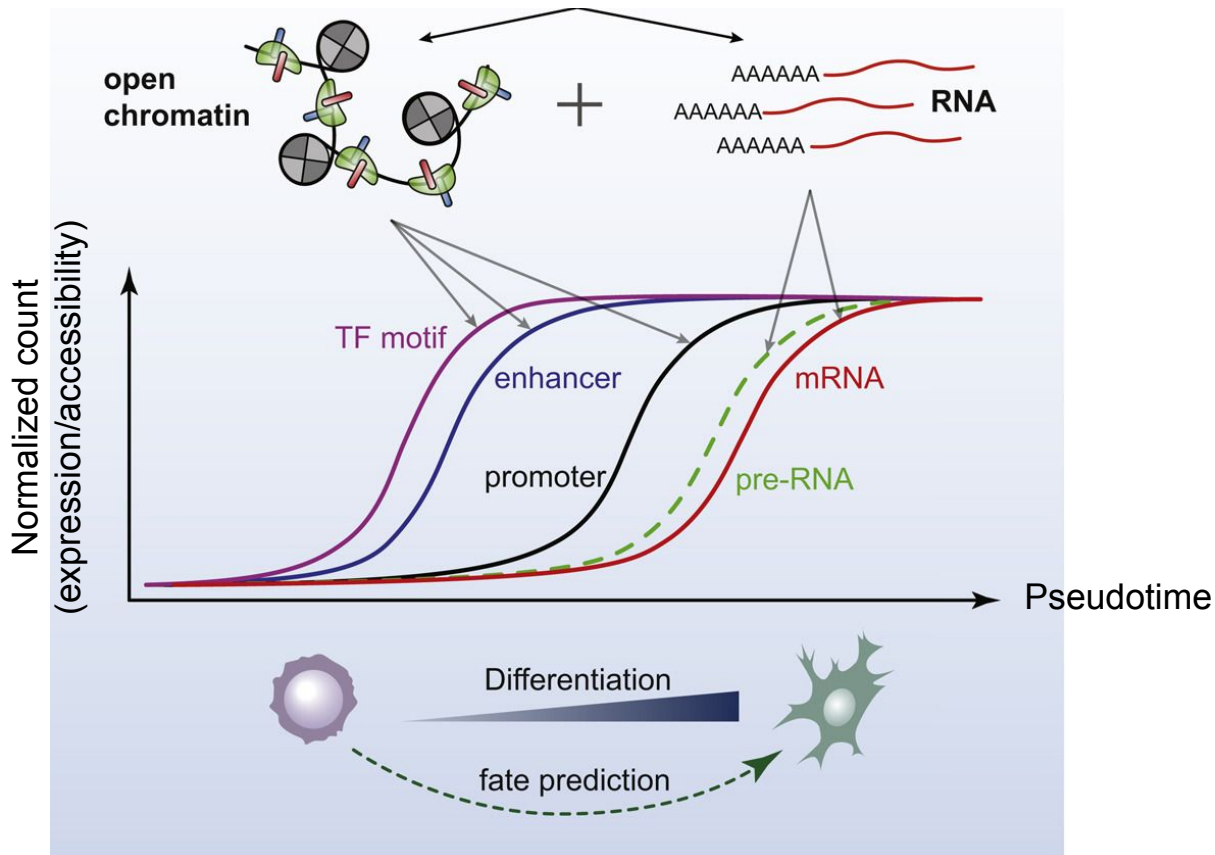
The first weak spot in single-cell multiome (RNA+ATAC) integration is the peak-linking problem

Many GRN-inference methods rely on correlations between gene expression and the accessibility of peaks to link peaks to target genes.

- No matter the method, this is very inaccurate. In particular, many true peak-gene links do not show a strong correlation.
- If you have alternative sources (e.g. the ENCODE rE2G predictions), consider using that instead.
- If you must infer it from your data, make sure to use a method that incorporate distance-based information as well.

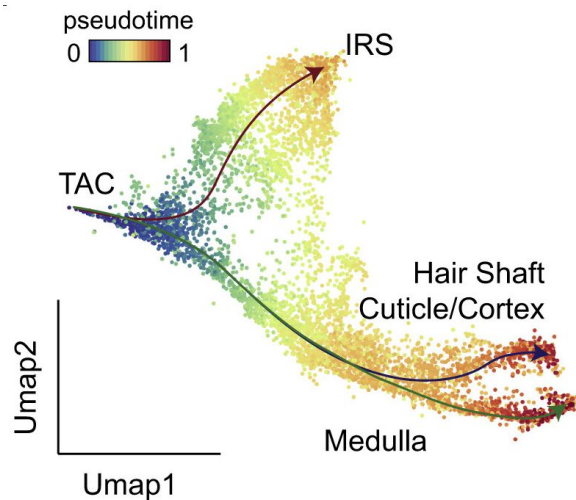
Second cautionary note on RNA + ATAC integration: the temporal dynamics are very different

During differentiation, changes in accessibility precede changes in gene expression

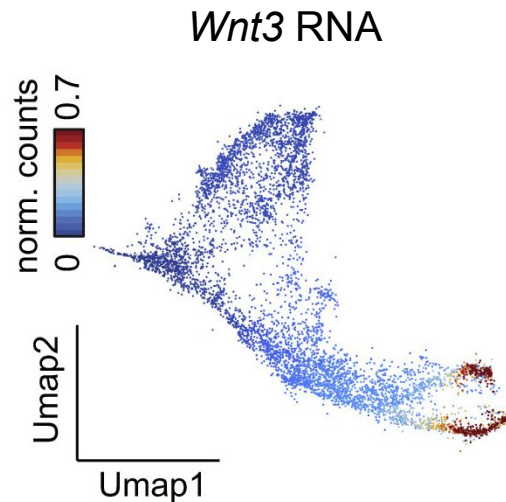
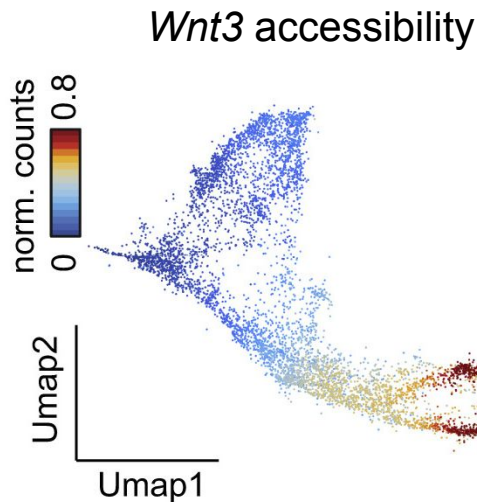


Second cautionary note on RNA + ATAC integration: the temporal dynamics are very different

During differentiation,
changes in accessibility
precede changes in gene
expression



Mouse hair follicle differentiation



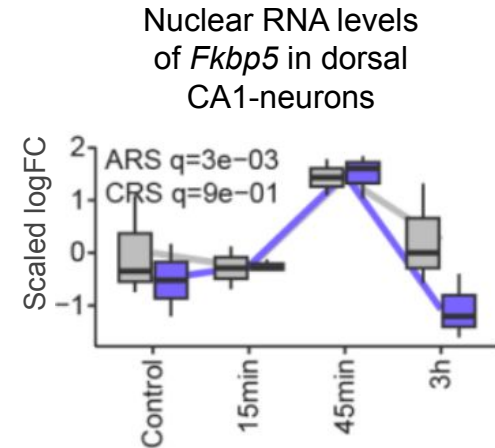
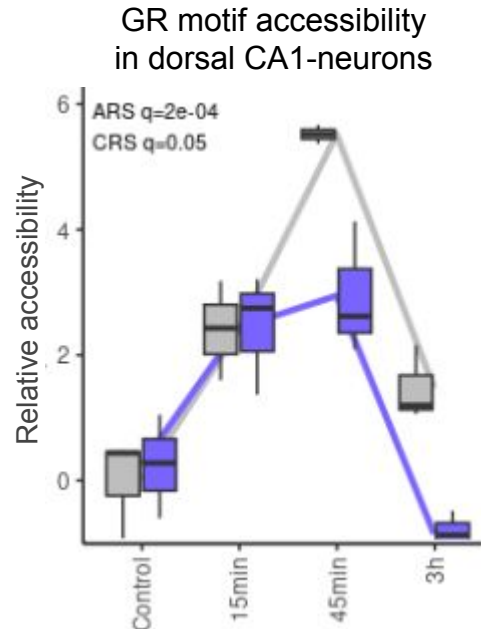
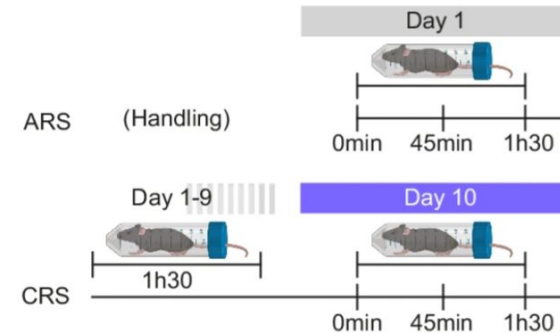
Example: glucocorticoid receptor (GR) activation upon stress

We used single-cell RNA and ATAC to monitor changes induced in the hippocampus by acute restraint stress (ARS) in naive mice and mice with a history of ARS exposure.

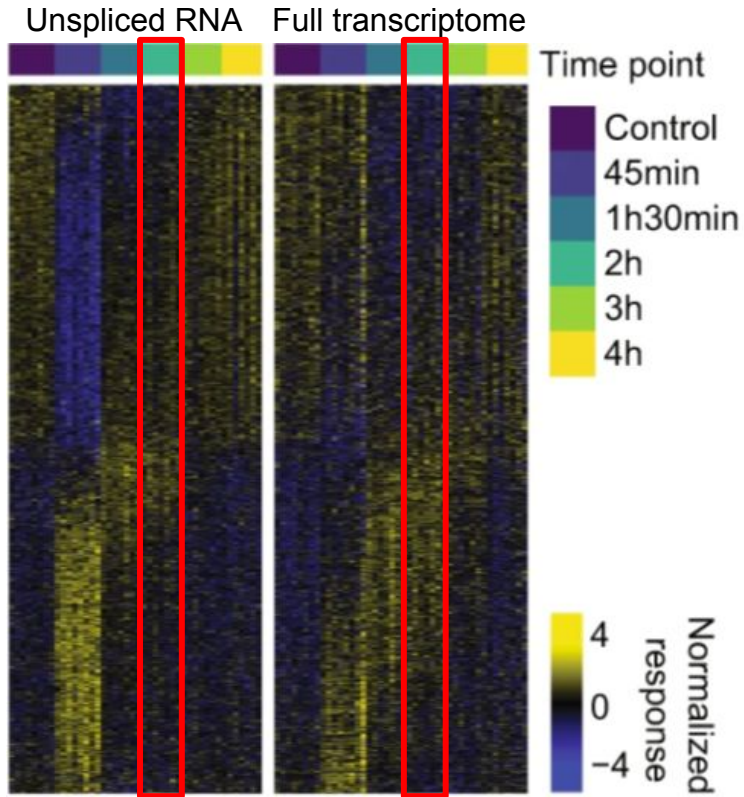
Glucocorticoids are released in the bloodstream within minutes of stress, and reach the brain in ~10min.

At 15min, we see a strong increase in GR motif accessibility.

However, we do not yet see a response in the nuclear RNA levels of its targets (e.g. here *Fkbp5*).



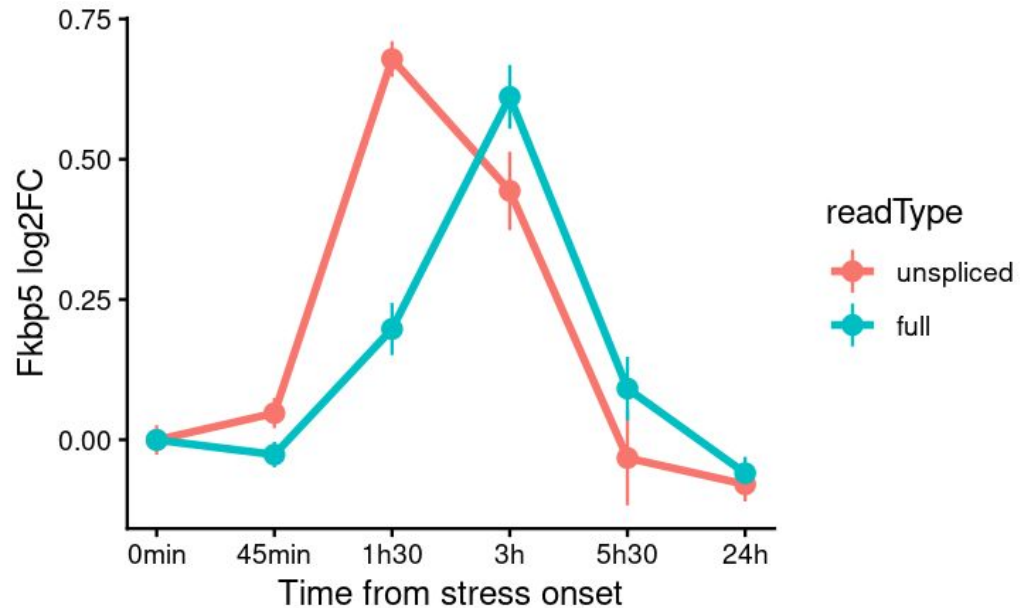
The transcriptome of a cell does not reflect *current* transcription, but what has transpired in the last hour



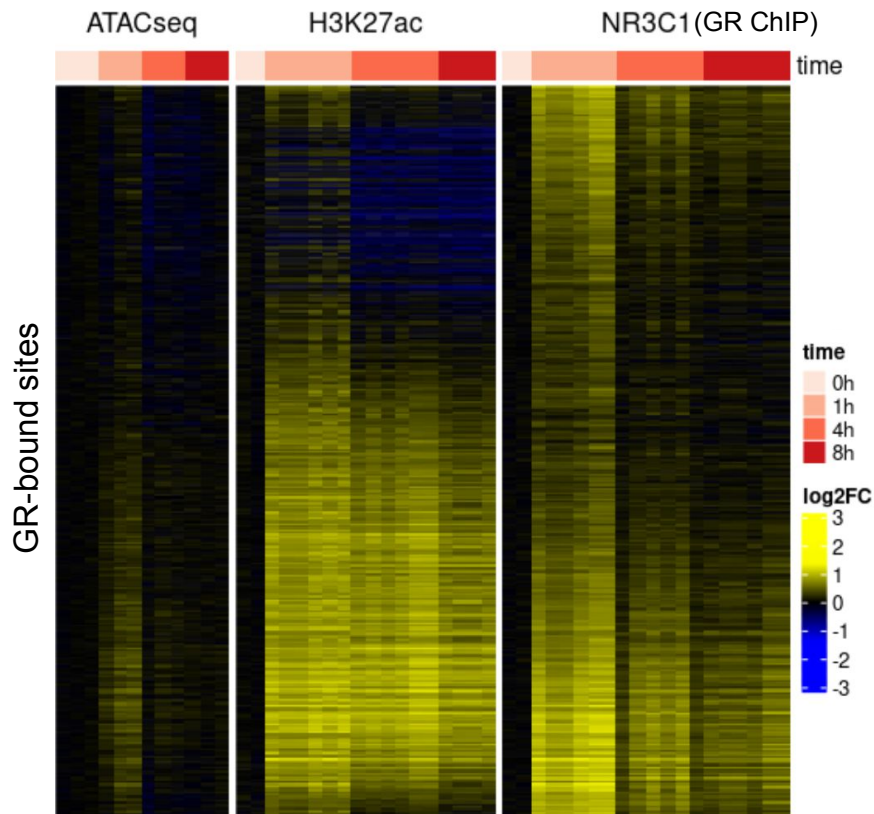
(Adapted from von Ziegler et al., Nat Comm 2025)

PRO-seq and GRO-seq capture nascent transcription.

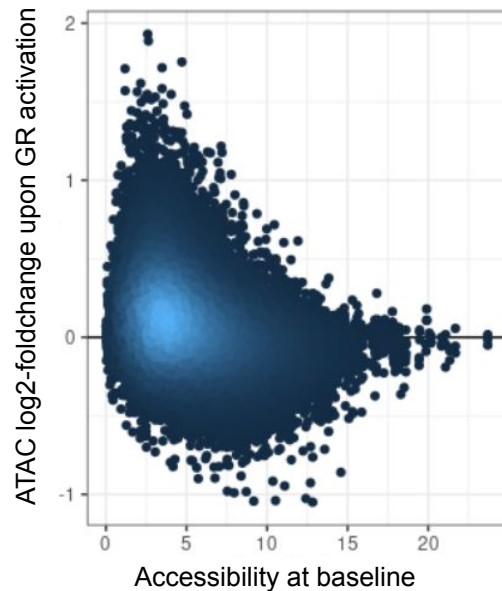
In standard bulk RNAseq, unspliced reads can be used as a proxy for more recent transcription



Third cautionary note on RNA + ATAC integration: epigenomic signals tend to plateau



Effect of GR activation in vitro across sites gaining GR binding



(Reanalysis of data from McDowell et al., Genome Research 2018)

Getting started with multi-omic integration

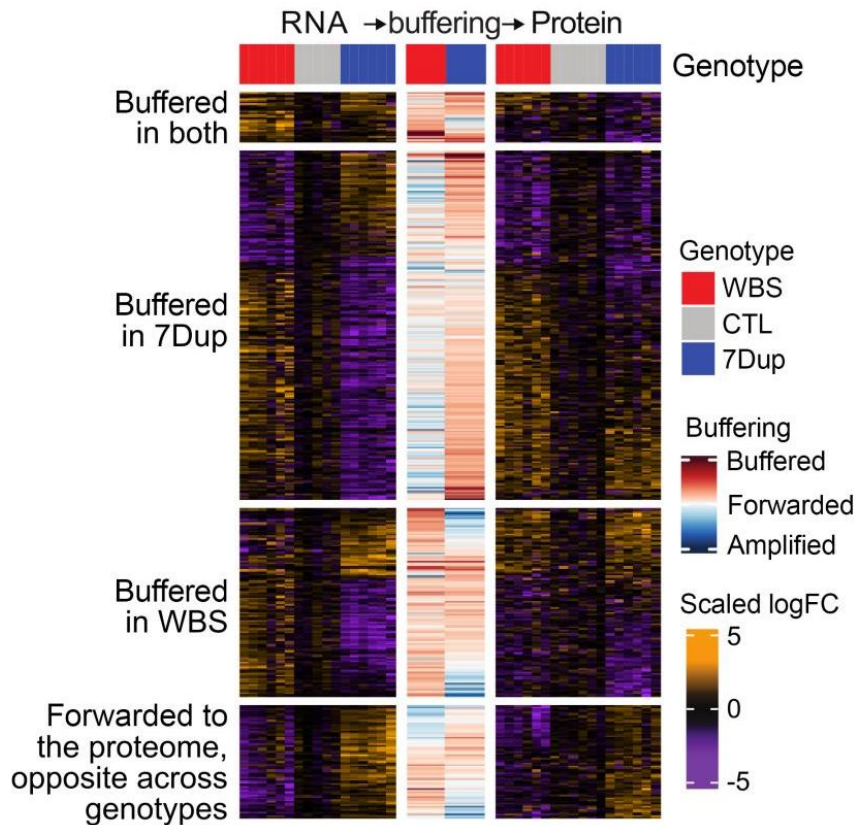
Assuming that you can link features across modalities : cluster features based on the (relative) multi-omic data, and then characterize the clusters.

Pro:

- Simple, not much bias/assumption
- Excellent for exploration (i.e. when no pre-defined hypothesis to test)

Con:

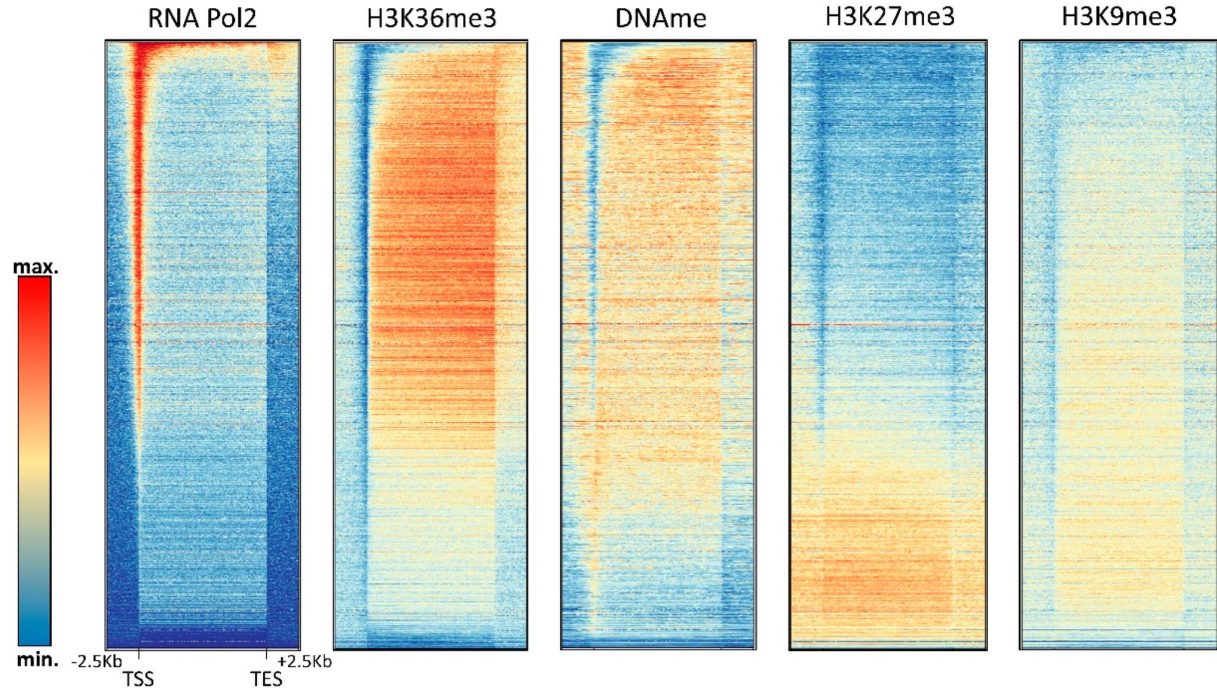
- Clustering not always trivial
- Effects in few features can get lost in the broader patterns



(Adapted from Mihailovich et al., JCI 2024)

Depending on the type of signal being studied, this can also be done with spatial signals

This is typically useful for epigenetic modifications, less so for TFs



(Jeltsch et al., Genes 2018;
see also Gu et al., BMC Genomics 2018, or our lecture on DNA methylation)

Multi-omics factorization (MOFA+)

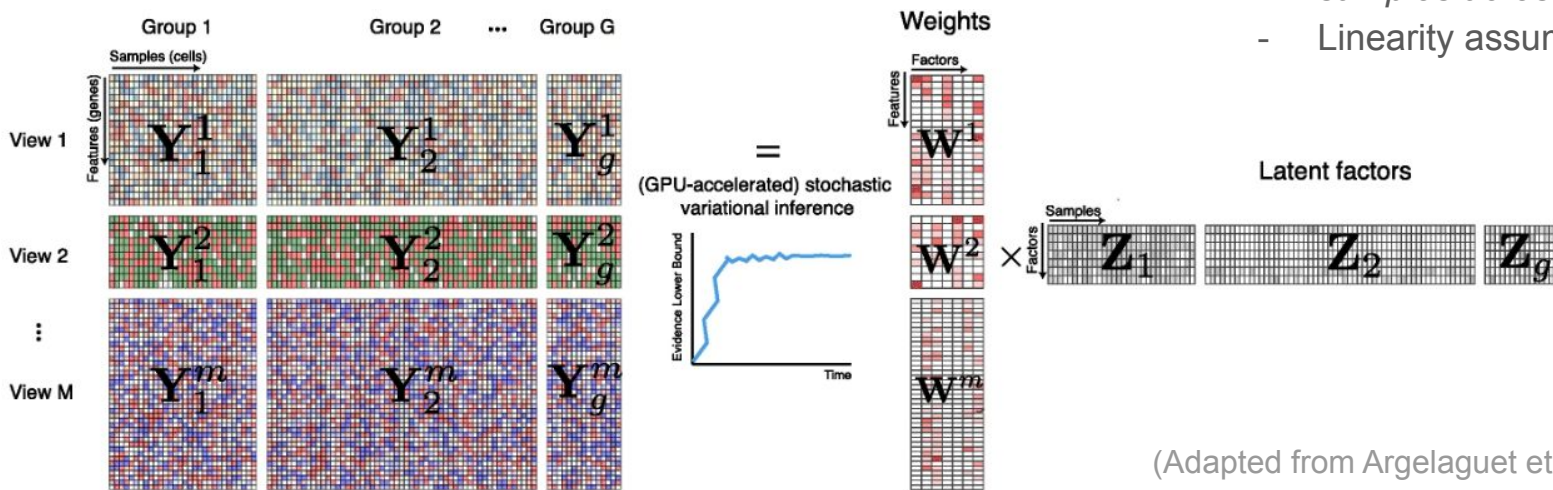
MOFA+ learns latent factors that can explain variance (across samples) across modalities. These factors can be associated to biological conditions, and their loadings on features can be used for enrichment analysis (GSEA).

Pro:

- does not require pre-defined linkage between features

Cons:

- Requires decent amount of samples (i.e. >20), and *the same samples across modalities*
- Linearity assumptions



(Adapted from Argelaguet et al., Genome Bio 2020)

Course project

Grading and expectations

- 50% of the grade is based on **weekly exercises**
 - Exercises should be **submitted via github**, by thursday noon the following week
 - The best half of the exercises will make up the grade
- 50% of the grade is based on the **project** (groups of 2-4 persons)
 - The project can be either:
 - Re-producing the analyses from a publication (in a critical fashion)
 - Analyzing new data (e.g. yours or in collaboration with a group)
 - The project *must be discussed and approved in advance*
 - The expected outputs of the project are:
 - a report (e.g. ~10-15 pages) with embedded full code and figures, and including an introduction and discussion of the results
 - Deadline: before the end of the day on **July 3rd**

Evaluation scheme for the course project

- Format and formal requirements (1/6)
 - Rendered markdown (html or pdf), proper scientific references, figure legends, etc.
- Introduction/conclusion (1/6)
 - More is not necessarily better: ask yourself what background would your fellow students need to understand your problem, analyses and observations
- Analysis
 - **Correctness**, i.e. lack of mistakes (1/6)
 - **Adaptability/creativity**, i.e. whether you could adapt (e.g. the tools seen in class) to your purposes (1/6)
 - **Appropriateness**, i.e. whether you used the right tool/visualization to address a question (1/6)
 - **Interpretation/discussion**, i.e. whether you correctly describe and interpret your figures and analysis results (1/6)
- Extra 1/6 (summing to 7 of max 6 points) for difficulty / going beyond the expectations (to not penalize those who undertook something harder)

A few (optional) tips for preparing your project report

- It's often easier to split a project into different markdown documents for different steps
 - e.g. one that downloads the data, another processes it, another that answers specific questions or makes the figures with which you'll tell your story
 - you may want to number the documents so that they appear in the right order
 - if you want to go deeper, [workflow](#) is a great markdown organization and versioning framework
- If you're wondering how to do something in rmarkdown, consult this [online book](#)
 - See especially:
 - code [chunk options](#)
 - the [section about references and bibliography](#)
(most reference management software, e.g. Zotero and the likes, can export references in the *bibtex* format required by rmarkdown)

Assignment

Come up with a preliminary plan for your project, summarizing:

1. What is the topic?
2. What data will you be using?
 - Be precise and concrete:
 - Experimental design and sample size
 - Where is the data available, at which processing step, etc?
3. What are the analyses you wish to reproduce, or the questions you wish to answer?

Write that up in a Rmarkdown that you can render and upload to your repository.

This is not a final plan, but the start of a discussion – make the best use of it !